

DC – AC Coefficients based Multiple Watermarking Technique

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Abstract— DC coefficient based watermarking schemes are known for good robustness against common image processing operations. This paper use scheme based on Quantization Index Modulation (QIM) in which both DC and AC coefficients are quantized for watermark embedding. The purpose is to increase the embedding capacity while maintaining constant robustness. Step size of quantizer decides robustness in QIM based methods. For this work, step size is decided based on the frequency content of an image. It results in increased robustness for fixed capacity or vice versa.

Keywords- Data Hiding; DC-AC coefficient; DCT; QIM; Watermarking.

I. INTRODUCTION (HEADING 1)

Using DC coefficients for embedding watermark is popular because resulting method will be robust. It is due to the fact that 1) Popular compression algorithms do not affect the DC coefficient. 2) DC coefficients have very large energy.

Traditionally DC coefficients based methods are known to have limited embedding capacity and poor perceptual quality for watermarked image. However if the parameters are properly adjusted these conflicting requirements can be optimized. Authors in [1] have presents a DC coefficient based embedding scheme. They discussed relationship between the embeddable location and noise caused by embedding watermark at that location. In [2] authors studied different characteristics of DC coefficients. They concluded that embedding in DC coefficients gives better performance than middle or low frequency embedding. Authors in [3] used AC and DC coefficients for embedding in audio. Watermark is embedded in DC coefficients and 4th AC coefficients alternatively. Authors claims to have obtained good compromise between watermark recovery and audible quality. Usage of Reed Solomon (RS) code for DCT coefficients using QIM was shown in [4]. Using an X-OR-ing function authors showed complete watermark recovery.

Methods discussed above used a fixed step size during watermark embedding. This results in **analogous** performance for all images in spite of the fact that they are derived from different distributions. Method in this paper uses different step size for each image. The optimization between the robustness and embedding capacity can be achieved if the quantizer step

size is adapted to image texture. Thus one can increase robustness for fixed capacity by increasing step size.

The method improves capacity by hiding in DC and AC coefficients simultaneously. Most of prior works had embedded 4096 bits in 512 x 512 as against 8192 bits in the proposed work. This is done by embedding additional bits in AC coefficient which is strong enough to survive common image processing operations.

The proposed method is organized as follows; Section II describes method for deciding an adaptive step size. Section III covers watermark embedding and watermark extraction. Section IV covers the experimentation and results while conclusion and discussion are drawn in section V.

II. DECIDING STEP SIZE

It is well known fact in the domain of data hiding that image with higher texture activity can hide large number of bits without significant degradation in perceptual quality. It is difficult for Human Visual System (HVS) to detect the changes in the region of high texture. Using this aspect first image shown in Figure 1 can hide about 28000 bits and second one can hide about 7000 bits using QIM based embedding for PSNR > 50 dB.

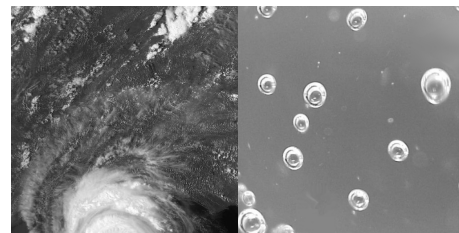


Figure 1 Hurricane Andrew Bubbles

For image like Hurricane Andrew one can trade robustness for capacity and perceptual degradation by increasing the step size. In QIM increasing step size results in higher robustness at the cost of reduced perceptual quality. Reduced perceptual quality can be compensated well if image has high texture activities. This notion is used in current method to increase the step size if the image contains higher texture activity.

The question to be addressed is how to detect the images with high texture activity quantitatively. Initially entropy was

used as measure for detecting image with large texture changes. Table I indicates entropy for some images.

TABLE I IMAGE AND THEIR ENTROPY VALUE.

Image	Entropy
Bubble	5.56
Building	6.44
Cameraman	7.04
Hurricane Andrew	7.26
Lena	7.32
Mandrill	7.29
Pirate	7.23

It can be observed that entropy can act as a measure for texture within an image and distinguish images like Hurricane Andrew and bubble. Nevertheless when the entropy does not have a large spread among values it becomes difficult to segregate images. For e.g. image like Lena and Mandrill have very close entropy value by large difference in texture activity perceptually. Thus it becomes necessary to use other measure for deciding texture activity. Edge content within an image can be used to decide texture activity. High pass filtering is performed on image to extract edges. The resulting image is binary. The number of 1's in binary image is counted to find a measure called as "edge count". Higher value of edge count indicates large textural changes. Figure. 2 shows high pass filtered images bubble and Hurricane Andrew. Table II indicates the edge count for images.

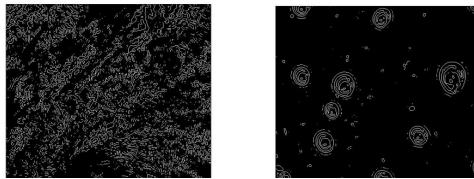


Figure 2 High pass filtered images

TABLE II EDGE COUNT FOR IMAGES

Image	Edge count
Bubble	5768
Building	14081
Cameraman	14930
Hurricane Andrew	24384
Lena	13865
Mandrill	31374
Pirate	20102

It can be observed from Table II that edge count covers a significantly large span of values. High value of edge count indicates good texture activity and vice versa. Method uses three different values of step size based on the edge count. They are used as follows in Table III:

TABLE III EDGE COUNT AND STEP SIZE

Range of edge count	Step size
< 6000	10
6000 to 15000	20
> 15000	30

These values of step size are arrived by heuristic approach through experimentation for given value of PSNR. These values of step size will be used in watermark embedding and extraction described in next section.

III A WATERMARK EMBEDDING

- Divide the source image into non overlapping blocks of 8 x 8.
- Compute forward DCT for each of these blocks.
- Scan the coefficients in zigzag fashion.
- DC coefficient and 1st AC coefficient are used for watermark embedding in block. These locations are shared between encoder and decoder. The other coefficients in block are left unaltered.
- 8192 random watermark bit ('0' and '1') pattern with zero mean, unit variance is generated.
- Watermark is embedded by rounding coefficients to even value if '0' is embedded and to odd value if '1' is embedded.
- Step size obtained as per edge count from Table 3 is used for QIM embedder.
- Apply inverse DCT to obtain watermarked image

III B WATERMARK DECODING

- Decoder shares locations of potential coefficients in 8 x 8 blocks. The decoding algorithm is as follows
- Divide the received watermarked image into non overlapping 8 x 8 blocks.
- Compute forward DCT for each of these blocks.
- Scan the coefficients in zigzag fashion.
- DC coefficient and 1st AC coefficient are used for watermark detection.
- These coefficients are rounded to nearest integer level. If the integer is even it is decoded as '0'. If integer is odd then it is decoded as '1'.
- The extracted bit pattern can be compared with original one to derive the method statistics

IV EXPERIMENTATION AND RESULTS

The algorithm was applied to 8 bit gray scale images with size 512 x 512. Figure3 shows the test images which were so chosen to represent different distributions.

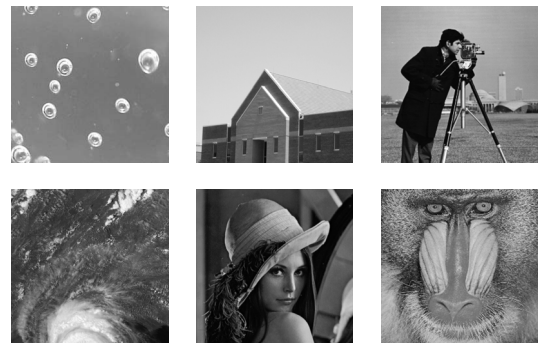


Figure 3 Test Images

Various test parameters at the encoder and the decoder are computed to evaluate the performance of scheme. The parameters evaluated at encoder are between original and watermarked image. They are:

1. Similarity factor (SF) or Normalized correlation (NC).
2. Root mean square error (RMSE).
3. Document to watermark ratio (DWR in dB).
4. Peak Signal to Noise ratio (PSNR in dB)

For comparison with other method, parameter listed in Table 4 are for derived for image Lena with step size of 20 and number of bits embedded were 8192.

TABLE IV ENCODER PARAMETERS

NC	RMSE	DWR	PSNR
0.99	1.44	46.59	51.72

Various parameters evaluated at decoder are as follows.

1. Watermark to Noise Ratio (WNR in dB)
2. Normalized correlation (NC).
3. Number of Errors
4. Bit error rate (BER)
5. Signal to Noise Ratio (SNR in dB)
6. Root mean square error (RMSE).
7. Peak Signal to Noise ratio (PSNR in dB)

Parameters 1 to 4 are between recovered and original watermark. Parameters 5 to 7 are for watermarked image which enters communication channel, gets corrupted by noise within and appears out of it. Table V shows various decoder parameters for image Lena with 8192 bits recovered with various attacks.

TABLE V DECODER PARAMETERS

JPEG compression with quality factor of 50						
WNR	NC	Bits error	BER	SNR	RMSE	PSNR
58.3408	0.99	12	0.0015	31.76	3.03	44.31
Gaussian Noise $N(0,0.0005)$						
17.59	0.91	706	0.086	25.74	4.10	41.30
Salt and pepper Noise (0.001)						
28.5093	0.97	237	0.028	46.07	1.48	51.46
Median Filter [3 x 3]						
23.1017	0.9504	407	0.0497	32.63	2.90	44.74

Table VI shows various decoder parameters for image Hurricane Andrew with 8192 bits embedded, JPEG compression with Q=50 and step size of 30. This will increase the robustness with reduction in PSNR value as indicated in Table VI.

TABLE VI DECODER PARAMETERS FOR HURRICANE ANDREW WITH STEP SIZE 30

WNR	NC	Number of bits in error	BER	SNR	RMSE	PSNR
Inf	1	0	0	22.36	5.20	38.90

In Table VII, comparison of proposed work with other methods is shown in terms of PSNR and NC. JPEG compression with Q=50 and step size of 20 is used as a bench mark for comparison among these methods.

TABLE VII COMPARISON AMONG DIFFERENT METHOD WITH THE PROPOSED METHOD

Method →	[1] Gaorong Zeng et.al.	[2] Jun Xiao et.al.	Proposed method
PSNR	38.29	38	44.31
NC	0.95	0.94	0.99
Number of bits embedded	4096	4096	8192

In terms of other attack like median filtering with size of 3 x 3 the comparison is illustrated in Table VIII.

TABLE VIII COMPARISON FOR MEDIAN FILTERING 3 X 3

Method →	[1] Gaorong Zeng et.al.	Proposed method
PSNR	38.6	44.74
NC	0.8913	0.95
Number of bits embedded	4096	8192

Effect of JPEG compression on watermark recovery and other parameters is studied by subjecting watermarked image to JPEG compression with Q varying from 0 to 100. Figure 4a to 4d shows variation of Q on parameters like WNR, NC, BER, and PSNR

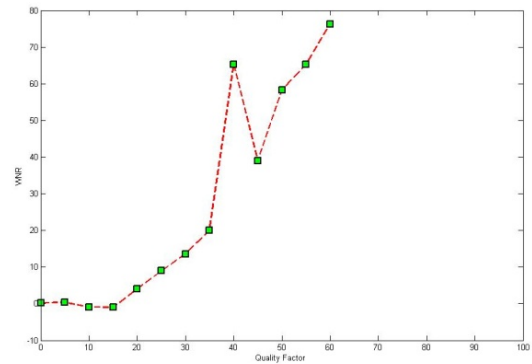


Figure 4a Q vs. WNR

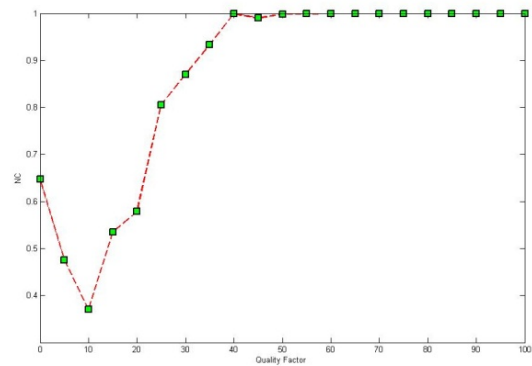


Figure 4b Q vs. NC

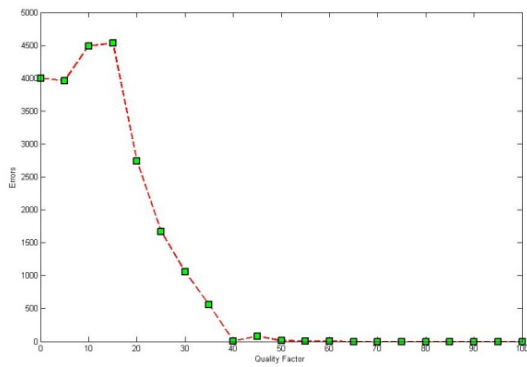


Figure 4c Q vs. BER

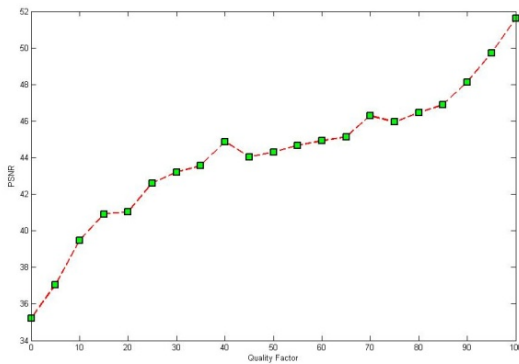


Figure 4d Q vs. PSNR

V CONCLUSION AND DISCUSSION

Method demonstrates improved robustness for superior capacity than other existing works. The step size is chosen based on the textural content of an image. This will lead to increased robustness with compensating perceptual quality. However decision on step size value for a given image is taken heuristically. Future work can look into deriving this value in more sophisticated mode.

ACKNOWLEDGEMENT

Test images for experimentation were used from website <http://www.imageprocessingplace.com>. I would also like to acknowledge help of Dr. Kaushal Solanki, Mayachitra Inc. U.S. for helping me in understanding and implementing concept of QIM.

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